

Cross-Dataset Evaluation of mBERT and MuRIL for Hinglish and Hindi-English Harmful Speech Detection

Working Paper Draft v0.5: 2026-10-08

This is an application research draft, not a published paper.

Author: Om Patnaik

Abstract

Hinglish and Hindi-English code-mixed harmful-speech detection is difficult because online text varies across language, script, platform, topic, and annotation policy. This project compares mBERT, a general multilingual BERT model, with MuRIL, an Indian-language-focused model, for binary harmful-speech classification across three dataset situations: a Kaggle Hinglish hate subset, a CM code-mixed offensive dataset, and THAR targeted religious hate. The strongest evidence comes from matched multi-seed experiments using seeds 7, 13, and 42: mBERT has higher observed Macro F1 than MuRIL on Kaggle Hinglish hate, mBERT has a small observed advantage on CM code-mixed/offensive data, and MuRIL has higher observed Macro F1 on THAR targeted religious hate. Cross-dataset, TF-IDF baseline, and mixed-training results show that label definition, platform, topic, and script mix affect model behavior under this setup. The experiments do not isolate whether script, platform, topic, or label policy causes each difference. The central claim is therefore conditional rather than universal: neither mBERT nor MuRIL is globally better for Hinglish harmful-speech detection under the current evidence.

1. Introduction

Hindi-English code-mixed text is common in Indian and South Asian online spaces. A single post may combine English, Romanized Hindi, Devanagari, slang, emojis, usernames, hashtags, and platform-specific context. This makes harmful-speech detection harder than ordinary binary text classification because the model must handle noisy multilingual input and unstable definitions of harm.

The project began with a direct question: whether MuRIL, because it is focused on Indian languages, would be better than mBERT for Hinglish hate-speech detection. As more datasets were added, the question became more interesting. The winner changed depending on the dataset. This shifted the project from a simple model comparison into a study of cross-dataset robustness.

The current research claim is not that mBERT is better in general or that MuRIL is better in general. The claim is that model ranking depends on dataset situation. A dataset situation includes the positive-label meaning, platform, topic/domain, script composition, and whether the model is tested in-domain or transferred from another dataset.

2. Research Question

The main research question is:

Does Indian-language-specific pretraining in MuRIL improve hate/offensive speech detection for Hinglish and Hindi-English code-mixed text compared with general multilingual pretraining in mBERT?

The current answer is conditional:

mBERT has higher observed scores on the matched Kaggle Hinglish hate condition.

mBERT has a small observed advantage on the matched CM code-mixed/offensive condition.

MuRIL has higher observed scores on the matched THAR targeted religious-hate condition.

Cross-dataset and mixed-training results show that label definition and dataset domain strongly affect model behavior.

This framing is meant to support a cautious application research paper: the contribution is not a new model architecture, but a careful comparison showing that evaluation conditions can reverse model rankings.

3. Related Work / Background

BERT introduced deep bidirectional Transformer pretraining for language understanding tasks and provides the base architecture behind many later text classifiers [1]. mBERT extends this style of pretraining to many languages and is commonly used as a multilingual baseline.

MuRIL was developed for Indian-language representation learning and includes Indian-language and transliterated signals that make it a natural candidate for Hinglish and Hindi-English code-mixed classification [2]. The comparison between mBERT and MuRIL is therefore meaningful because Hinglish sits between English, Hindi, Romanized Hindi, and Indian social media usage.

Prior Hindi-English code-mixed hate-speech work, including Bohra et al., shows that code-mixed harmful-speech detection cannot be treated as ordinary English hate-speech detection [3]. This project builds on that idea but focuses specifically on whether the relative ranking of mBERT and MuRIL remains stable across datasets.

The dataset sources used here include the Kaggle Code-Mixed Hinglish Hate Speech Detection Dataset [4], the CM hate-speech repository [5], and the THAR targeted religious-hate dataset [6; 7]. Public source checks on 2026-10-04 found that Kaggle metadata lists the Kaggle dataset as MIT, while GitHub reports no detected license for the CM and THAR repositories. For that reason, this paper reports derived experimental results and citations, but the repository does not redistribute raw CM or THAR data.

4. Datasets

The project uses three primary source-backed datasets. The 79-row benchmark is retained only as a diagnostic probe and excluded from primary claims.

Dataset ID	Rows	Positive label	Platform/domain	Script profile	Paper role
kaggle_hinglish_hate	4,780	hate	mixed/unclear from local metadata	100% Latin script in processed subset	primary matched Hinglish hate condition
cm_splits_codemixed	3,900	offensive	Indian politics / Twitter/X	mostly Latin with some mixed Latin-Devanagari	external code-mixed offensive condition
thar_religion	11,549	AntiReligion	YouTube religious comments	mostly Latin, meaningful Devanagari portion	targeted religious-hate condition
existing_79_row_benchmark	79	hate	unclear/manual/artificial	100% Latin script	diagnostic only

4.1 Kaggle Hinglish Hate

The Kaggle dataset is used as the closest dataset to the original project aim: Hinglish hate/non-hate classification. The local processed file contains 4,780 Hinglish rows, with 2,914 negative examples and 1,866 positive examples. Its processed subset is 100% Latin script. This matters because MuRIL's Indian-language pretraining may not automatically help when the task is heavily Romanized and resembles Latin-script social media.

4.2 CM Code-Mixed Offensive

The CM dataset contains Indian politics and social-media text from Twitter/X. Its project label is mapped from offense, so its positive class is hate-adjacent rather than strict hate speech. The processed file contains 3,900 rows, with 2,455 negatives and 1,445 positives. This dataset is useful because it adds a strong Indian-context code-mixed condition, but it should not be treated as semantically identical to strict hate speech.

4.3 THAR Targeted Religious Hate

THAR is a targeted religious-hate dataset from YouTube comments. Its positive class is AntiReligion, not general hate or general offensive language. The processed file contains 11,549 rows, with 6,095 negatives and 5,454 positives. THAR is narrower than the other datasets, but that narrowness is methodologically useful because it tests whether MuRIL has higher observed scores in a targeted Indian-language harmful-speech situation.

4.4 Diagnostic 79-Row Probe

The 79-row benchmark is excluded from primary results because its provenance is uncertain, it may be manually written or AI-generated, it contains duplicate rows, and its labels may not consistently represent targeted hate. It remains useful as a lesson: small unclear benchmarks can produce unstable model rankings and misleading confidence.

5. Models And Baselines

The transformer models are:

mBERT: bert-base-multilingual-cased

MuRIL: google/muril-base-cased

Both are fine-tuned as binary sequence classifiers. The project also includes TF-IDF baselines:

TF-IDF with Logistic Regression

TF-IDF with Linear SVM

The baselines matter because harmful-speech datasets often contain strong lexical cues: repeated slurs, slogans, target-group terms, political phrases, and platform-specific patterns. If a TF-IDF model performs competitively, then transformer gains should not be exaggerated.

6. Methodology

The experiments use binary labels (0 and 1) across all datasets, but the meaning of 1 differs. This is intentional and documented: the project tests robustness across harmful-speech dataset situations, not one perfectly harmonized hate-speech ontology.

Transformer training used a controlled first-pass configuration:

Setting	Value
Epochs	2
Learning rate	2e-5
Maximum sequence length	128
Batch size on Mac MPS	8
Matched multi-seed seeds	7, 13, 42

Two epochs were used as a controlled first-pass setting across model/dataset combinations. The project prioritized comparable coverage across matched, cross-dataset, and mixed-training conditions over extensive hyperparameter tuning. The training script evaluates and saves the best epoch using Macro F1 on the same matched evaluation split reported in the tables. Therefore, the matched numbers are selection-set performance rather than untouched final-test estimates. For Kaggle and THAR, each seed controls both training randomness and the stratified 80/20 split; mBERT and MuRIL use the same split within each seed. CM combines the source train and val splits for training (3,476 rows), and uses the source test split (424 rows) for epoch selection and reported evaluation. The CM split stays fixed across seeds. A separate final test set or nested validation protocol would strengthen generalization estimates.

The primary metric is Macro F1 because it averages performance across both classes and is less misleading than accuracy under class imbalance. Positive recall is also tracked because false negatives are harmful examples that the model misses. In harmful-speech detection, a high false-negative rate means the system is too hesitant to flag harmful content.

7. Results

7.1 Matched Multi-Seed Results

The matched multi-seed results are the strongest evidence in the repository because they repeat the primary matched comparison across seeds 7, 13, and 42. They are still selection-set results because the reported evaluation split also guided best-epoch choice.

Dataset	Higher observed Macro F1	mBERT Macro F1	MuRIL Macro F1	mBERT positive recall	MuRIL positive recall	Interpretation
kaggle_hinglish_hate	mBERT	67.5 +/- 2.1	58.1 +/- 5.7	46.9 +/- 8.5	25.1 +/- 8.5	mBERT has a clear matched advantage; MuRIL misses many positives
cm_splits_codemixed	mBERT, narrowly	77.7 +/- 1.9	76.1 +/- 2.3	70.7 +/- 1.3	64.4 +/- 8.3	both are competitive; mBERT has a small observed advantage
thar_religion	MuRIL	74.7 +/- 0.1	76.5 +/- 1.3	79.3 +/- 2.0	79.7 +/- 0.6	MuRIL has the higher observed score on targeted religious hate

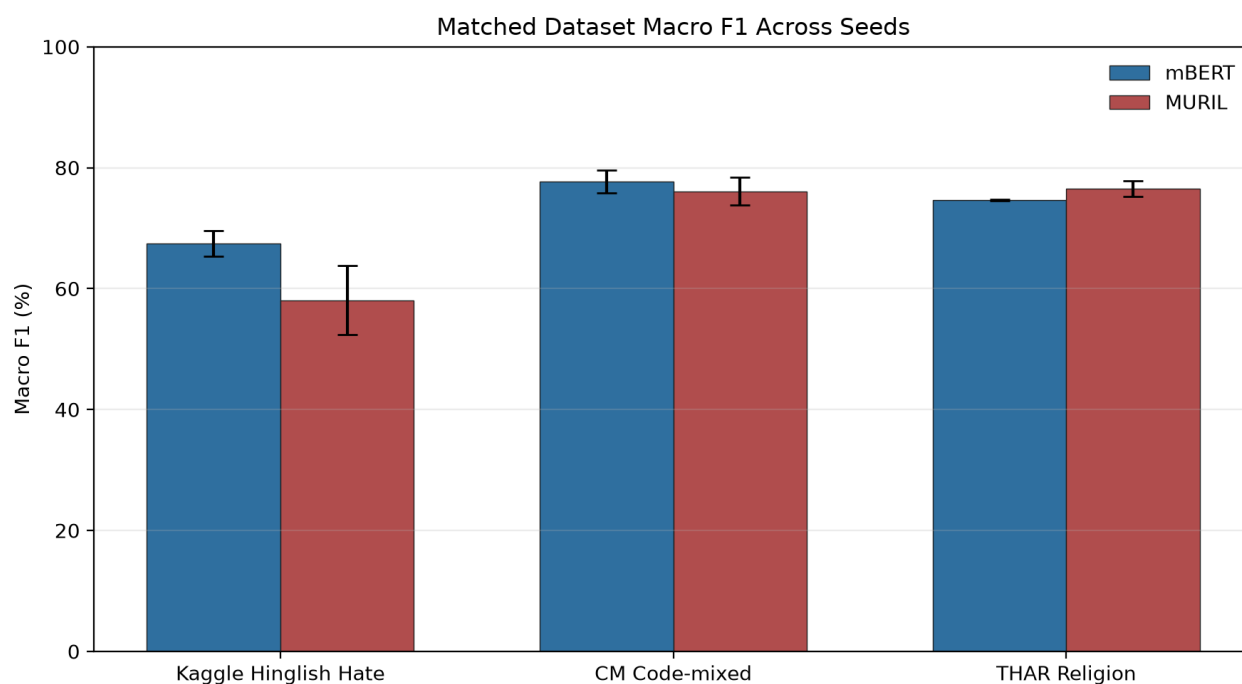


Figure 1. Matched selection-set Macro F1: mean and sample standard deviation across seeds 7, 13, and 42.

These results do not support a universal winner. They show that mBERT has higher observed scores on the Latin-script-heavy Kaggle condition and a small observed advantage on CM, while MuRIL has higher observed scores on targeted religious hate. For Kaggle and THAR, different seeds also change the stratified split membership, so the standard deviations combine training randomness and split variation. CM uses the fixed source split. The CM and THAR gaps are small relative to the limited seed count, and this draft does not report confidence intervals or statistical significance tests. They should therefore be treated as observed differences under this setup, not as established reliable superiority.

7.2 Cross-Dataset Results

The cross-dataset results are mostly single-seed evidence and should be treated as weaker than the matched multi-seed table. They are still important because they test whether a model trained on one dataset situation works on another.

The largest generalization gaps occur when models are trained on one positive-label definition and evaluated on another. Examples from the result analysis include:

Model	Train dataset	Test dataset	Matched Macro F1	Transfer Macro F1	Gap
MuRIL	thar_religion	kaggle_hinglish_hate	77.9	46.2	31.7
mBERT	thar_religion	kaggle_hinglish_hate	74.8	45.5	29.3
mBERT	cm_splits_codemixed	kaggle_hinglish_hate	78.3	51.6	26.6
mBERT	kaggle_hinglish_hate	thar_religion	65.6	44.8	20.8

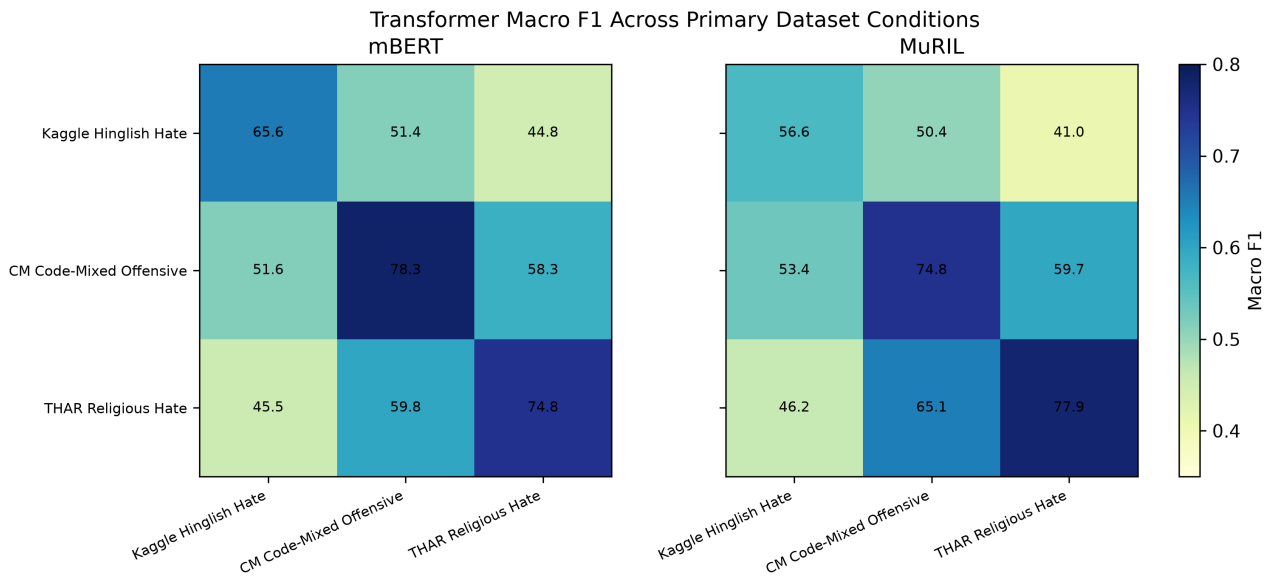


Figure 2. Exploratory single-seed (42) transformer Macro F1 across primary training and evaluation conditions.

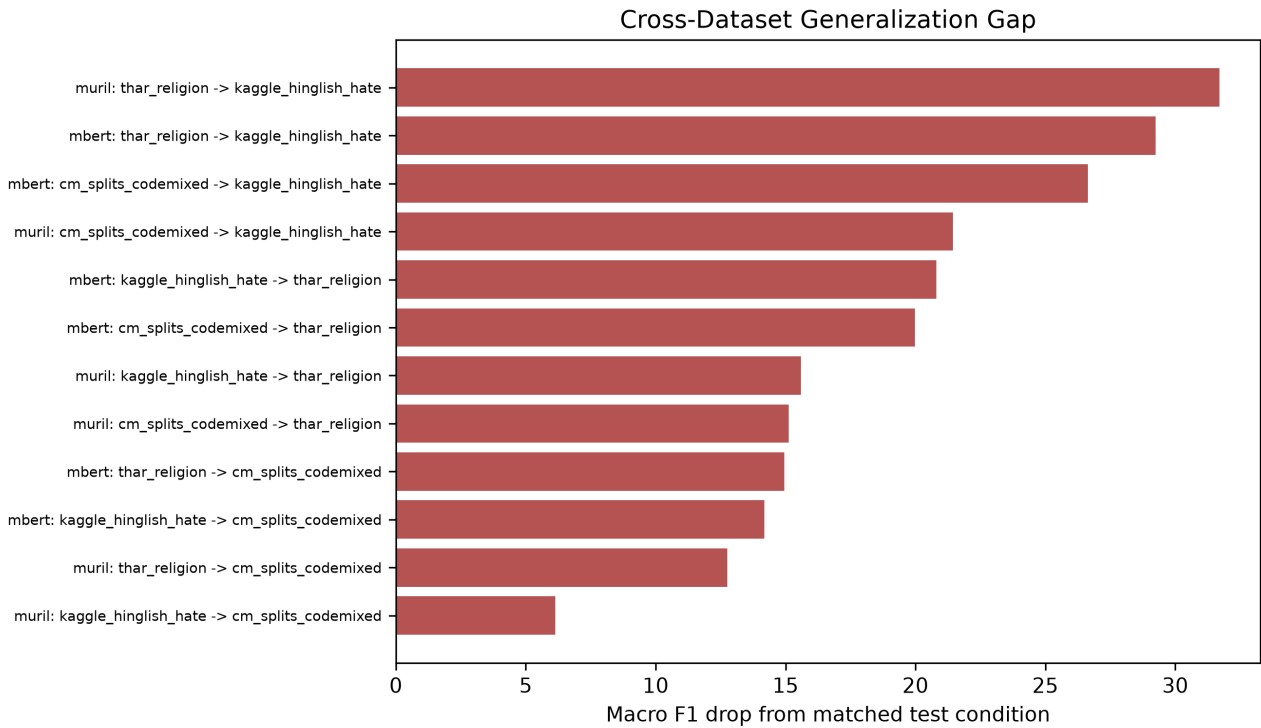


Figure 3. Exploratory single-seed (42) generalization gaps relative to the corresponding matched score.

This is the main evidence that cross-dataset robustness is weak. THAR-trained models do not become general Hinglish hate detectors, and Kaggle-trained models do not transfer cleanly to targeted religious hate.

The cross-dataset rows in docs/result_analysis_report.md use the earlier seed-42 matched/cross-dataset matrix. They are described here as single-seed transfer evidence unless rerun under the multi-seed harness.

7.3 TF-IDF Baseline Comparison

TF-IDF baselines are competitive in several settings. The following comparison uses the earlier exploratory single-seed (42) transformer matrix, rather than the three-seed means:

Train dataset	Test dataset	Best transformer Macro F1	Best TF-IDF Macro F1	Transformer minus baseline
cm_splits_codemixed	cm_splits_codemixed	78.3	77.6	+0.7

Train dataset	Test dataset	Best transformer Macro F1	Best TF-IDF Macro F1	Transformer minus baseline
kaggle_hinglish_hate	thar_religion	44.8	49.9	-5.2
thar_religion	kaggle_hinglish_hate	46.2	51.0	-4.9
thar_religion	thar_religion	77.9	70.6	+7.3

These results matter because they prevent overclaiming. Transformers are strongest in matched conditions, but simple lexical models can beat them in some transfer settings. This suggests that dataset-specific lexical cues, slogans, and target terms play a large role.

7.4 Mixed-Dataset Training Results

Mixed-training results are currently single-seed evidence. They should be presented as exploratory robustness tests, not as final settled claims.

Mixed train condition	Main mBERT behavior	Main MuRIL behavior	Interpretation
Kaggle + CM	mBERT improves Kaggle over Kaggle-only seed-42 and remains usable on CM, but weak on THAR	MuRIL collapses to all-negative predictions	related datasets can still produce model-specific instability
CM + THAR	mBERT remains strong on CM and THAR, weak on Kaggle	MuRIL collapses to all-negative predictions	MuRIL collapse is serious under this mixture
Kaggle + THAR	both models learn; mBERT has a small observed advantage on Kaggle, MuRIL has higher observed scores on THAR and CM	MuRIL does not collapse	collapse is condition-dependent, not universal
Kaggle + CM + THAR	both models learn; mBERT leads Kaggle/CM, MuRIL leads THAR	MuRIL stable and strong on THAR	broader mixing helps but does not solve robustness

The key result is not simply that mixed data helps. Mixed data can help, hurt, or destabilize training depending on source balance and label compatibility.

8. Error Analysis

The quantitative error analysis measures prediction outcomes and feature associations in the earlier exploratory single-seed (42) matrix. Automated heuristic tagging provides candidate cues for later human review; it does not establish causes of errors.

Quantitative error analysis shows that:

Kaggle-trained models miss many THAR positives, so broad Hinglish hate training does not transfer cleanly to targeted religious hate.

THAR-trained models miss many Kaggle positives, so targeted religious hate training does not create a general Hinglish hate detector.

mBERT tends to have lower false-negative rates on matched Kaggle and CM conditions.

MuRIL tends to have lower false-negative rates on matched THAR and THAR-to-CM transfer.

Devanagari-only rows show high false-negative rates, which supports further script-specific analysis.

The script `scripts/first_pass_manual_error_coding.py` assigns tags automatically using train/test identities, keyword lists, script flags, and text length. Historical filenames and the `manual_reason` column retain their original names for traceability, but these outputs are automated heuristic tags, not independently reviewed manual annotations. The tag `cross_dataset_label_mismatch` is assigned to every row whose training and evaluation datasets differ. Its 191 rows (67.0% of the 285-row sample) therefore describe the sample's transfer composition and cannot establish that label mismatch caused 67.0% of errors. Other keyword and script tags identify possible cues rather than verified explanations.

The qualitative lesson is that hate, offensive, and AntiReligion labels are not interchangeable. A text may be offensive but not targeted hate, or religious in topic but not anti-religion hate. Models trained on one label policy often fail when evaluated under another.

No independently human-reviewed error examples are reported here. The automated tags support example selection for future review, while conclusions about failure rates rely on the quantitative predictions.

9. Discussion

The central result is conditional model ranking. mBERT has higher observed scores on the Kaggle and CM matched settings, while MuRIL has higher observed scores on THAR. This is not a contradiction to hide. It is the research finding.

One possible explanation is script/domain alignment. The processed Kaggle dataset is entirely Latin script, and CM is mostly Latin-script political/social media text. mBERT may handle these Latin-script, English-adjacent subword patterns better. MuRIL's Indian-language-focused pretraining does not automatically help when the task is heavily Romanized and broad. However, the current experiments do not isolate script from platform, topic, label definition, or source balance.

THAR is different. It is targeted religious hate, includes a meaningful Devanagari portion, and focuses on Indian religious group references. MuRIL's higher observed THAR scores may reflect better alignment with Indian-language cues and target-domain signals. However, this should remain a cautious interpretation, not a causal proof.

The mixed-training results deepen the story. MuRIL collapses under some mixtures but not others, especially recovering when THAR is included with Kaggle or all three datasets. This suggests that model behavior may depend on label compatibility, source balance, threshold/decision boundary behavior, and the training mixture, not only on model architecture.

The TF-IDF results show that lexical cues remain powerful. A transformer that beats another transformer by a few points should still be compared against lexical baselines, especially when cross-dataset transfer rewards simple keyword overlap.

10. Limitations

The main limitation is label comparability. hate, offensive, and AntiReligion are related harmful-speech labels, but they do not define the same task. This limits any universal claim about model superiority.

Other limitations:

Mixed-training and cross-dataset transformer results are mostly single-seed evidence.

Only the main matched settings currently have three-seed means and standard deviations.

Hyperparameters were controlled rather than extensively tuned.

No confidence intervals or bootstrap intervals have been added yet.

Threshold tuning and class weighting need further investigation, especially for MuRIL collapse conditions.

Kaggle metadata lists the Kaggle source as MIT, but CM and THAR repository licenses are not detected by GitHub; raw CM and THAR data should not be redistributed from this repository without additional permission review.

Reported matched scores are selection-set scores because the evaluation split was also used for best-epoch selection.

The Kaggle dataset's exact source metadata and Indian-context status need additional review.

The CM dataset includes duplicates and some duplicate-label conflicts.

The 79-row probe is excluded from primary conclusions because its provenance is uncertain.

Automated error tags are rule-based screening aids; they have not been validated by independent human annotation and do not establish error causes.

These limitations do not invalidate the project. They define the correct strength of the claim: this is an application research draft with meaningful evidence, not a final peer-reviewed conclusion.

11. Tools Note

AI tools were used for coding, debugging, and documentation assistance; the research direction, result interpretation, and final claims were reviewed and owned by Om Patnaik.

12. Conclusion

This project shows that the mBERT-vs-MuRIL comparison cannot be answered with a single score. In matched multi-seed evaluation, mBERT has higher observed scores on Kaggle Hinglish hate and a small observed advantage

on CM code-mixed/offensive data, while MuRIL has higher observed scores on THAR targeted religious hate. Cross-dataset and mixed-training experiments show that harmful-speech detection is shaped by dataset situation: label definition, platform, topic, script composition, and source balance.

The most defensible conclusion is that model choice and dataset definition must be studied together. For Hinglish and Hindi-English harmful-speech detection, cross-dataset robustness is weak, and that weakness is a core finding rather than a failure to hide.

References

Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL-HLT, 2019.

Khanuja, S., et al. MuRIL: Multilingual Representations for Indian Languages. arXiv:2103.10730, 2021.

Bohra, A., Vijay, D., Singh, V., Akhtar, S. S., and Shrivastava, M. A Dataset of Hindi-English Code-Mixed Social Media Text for Hate Speech Detection. W18-1105, 2018.

Dhekane, S. Code-Mixed Hinglish Hate Speech Detection Dataset. Kaggle. Public metadata checked 2026-10-04; license field: MIT.

cm-hate-speech-detection contributors. cm-hate-speech-detection GitHub repository. Public metadata checked 2026-10-04; GitHub detected no license.

Sharma, D., Singh, A., and Singh, V. K. THAR: Targeted Hate Speech Against Religion: A High-Quality Hindi-English Code-Mixed Dataset with the Application of Deep Learning Models for Automatic Detection. ACM Transactions on Asian and Low-Resource Language Information Processing, journal article, 2024. DOI: 10.1145/3653017.

THAR contributors. THAR GitHub repository. Public metadata checked 2026-10-04; GitHub detected no license.